Nelco® N4000-7
Nelco® N4000-7 SI®

CAF Resistant, Low-CTE FR-4 Epoxy Laminate & Prepreg

The Nelco® N4000-7 low-CTE epoxy laminate and prepreg system is a cost-effective solution for use in a broad range of multilayer PCB designs requiring low Z-axis expansion, excellent thermal stability and superior hole-wall integrity.

Key Features

Outstanding Z-axis, thermal stability and hole wall integrity
- Improved thermal, mechanical and chemical properties that meet or exceed those of many high-Tg materials
- Suitable for cost-sensitive, high-reliability applications

Automotive Testing
- Passes the stringent Q1000 thermal cycle tests of -40ºC to +125ºC for 1000 cycles

SI (Signal Integrity) option
- When used, SI glass provides enhanced electrical performance which can be used for high speed, low loss designs when needed

CAF* resistant
- The low Z-CTE and proven CAF resistance allow N4000-7 to be used in high reliability applications typically reserved for higher Tg materials

Cost-effective standard FR-4 processing
- Key processing parameters of drilling, desmear and lamination use standard low Tg FR-4 methods
- Accelerated press cycles and drilling when compared to high Tg materials
- 45 min press at 170ºC and 200-300 psi.

And Much More
- Vacuum laminated
- Available in a wide variety of constructions, copper weights and glass styles including standard copper, double treat and RTFOIL® laminate.
- Meets UL 94V-0 and IPC-4101/98 specifications*
- All Nelco® materials are RoHS compliant.

* material also meets the specifications of IPC-4101/24 unfilled slash sheet.

Applications

- Wireless Handsets
- Wireless Infrastructure
- Digital Broadband Multilayers
- Automotive
- Underhood Automotive
- Backplanes
- Fine-Line Multilayers
- Fine Pitch BGA Multilayers
- Direct Chip Attach

Global Availability

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Park's UL file number: E36295

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Nelco® N4000-7 and N4000-7 SI®

CAF Resistant, Low-CTE FR-4 Epoxy Laminate & Prepreg

Mechanical Properties	N4000-7	N4000-7 SI	U.S. Units	N4000-7	N4000-7 SI	Metric	Test Method
Peel Strength - 1 oz. (35 micron) Cu							
After Solder Float	7.5	7.5	lb / inch	1.31	1.31	N / mm	IPC-TM-650.2.4.8
At Elevated Temperature	8.1	8.1	lb / inch	1.42	1.42	N / mm	IPC-TM-650.2.4.8.2a
After Exposure to Process Solutions	9.0	9.0	lb / inch	1.58	1.58	N / mm	IPC-TM-650.2.4.8
X / Y CTE [-40°C to +125°C]	12 - 15	12 - 15	ppm / °C	12 - 15	12 - 15	ppm / °C	IPC-TM-650.2.4.41
Z Axis CTE Alpha 1 [50°C to Tg]	50	50	ppm / °C	50	50	ppm / °C	IPC-TM-650.2.4.24
Z Axis CTE Alpha 2 [Tg to 260°C]	270	270	ppm / °C	270	260	ppm / °C	IPC-TM-650.2.4.24
Z Axis Expansion [50°C to 260°C]	3.7	3.7	%	3.7	3.7	%	IPC-TM-650.2.4.24
Young's Modulus (X / Y)	3.5 / 2.8	TBD	psi x 10^6	24.1 / 19.3	TBD	GN / m^2	ASTM D3039
Poisson's Ratios (X / Y)	0.13 / 0.11	TBD		0.13 / 0.11	TBD		ASTM D3039
Thermal Conductivity	0.525	0.525	W / mK	0.525	0.525	W / mK	ASTM E1461
Specific Heat	1.22	1.22	J / gK	1.22	1.22	J / gK	ASTM E1461
Electrical Properties							
Dielectric Constant (50% resin content)							
@ 1 MHz (TFC / LCR Meter)	4.5	4.0		4.5	4.0		IPC-TM-650.2.5.5.3
@ 1 GHz (RF Impedance)	4.0	3.6		4.0	3.6		IPC-TM-650.2.5.5.9
@ 2.5 GHz (Stripline)	3.9	3.4		3.9	3.4		IPC-TM-650.2.5.5.5
@ 10 GHz (Stripline)	3.9	3.3		3.9	3.3		IPC-TM-650.2.5.5.5
Dissipation Factor (50% resin content)							
@ 1 MHz (TFC / LCR Meter)	0.018	0.016		0.018	0.016		IPC-TM-650.2.5.5.3
@ 2.5 GHz (Stripline)	0.017	0.015		0.017	0.015		IPC-TM-650.2.5.5.9
@ 10 GHz (Stripline)	0.017	0.016		0.017	0.016		IPC-TM-650.2.5.5.5
Volume Resistivity							
C - 96 / 35 / 90	10^8	10^8	MΩ - cm	10^8	10^8	MΩ - cm	IPC-TM-650.2.5.17.1
E - 24 / 125	10^7	10^7	MΩ - cm	10^7	10^7	MΩ - cm	IPC-TM-650.2.5.17.1
Surface Resistivity							
C - 96 / 35 / 90	10^8	10^8	MΩ	10^8	10^8	MΩ	IPC-TM-650.2.5.17.1
E - 24 / 125	10^7	10^7	MΩ	10^7	10^7	MΩ	IPC-TM-650.2.5.17.1
Electric Strength	1250	1250	V / mil	$4.3x10^4$	$4.3x10^4$	V / mm	IPC-TM-650.2.5.6.2
Dielectric Breakdown	>50	>50	kV	>50	>50	kV	IPC-TM-650.2.5.6
Arc Resistance	124	124	seconds	124	124	seconds	IPC-TM-650.2.5.1
Thermal Properties							
Glass Transition Temperature (Tg)							
DSC (°C)	155	155	°C	155	155	°C	IPC-TM-650.2.4.25c
TMA (°C)	150	150	°C	150	150	°C	IPC-TM-650.2.4.24c
DMA (°C) (Tan d Peak)	160	160	°C	160	160	°C	IPC-TM-650.2.4.24.3
Degradation Temp (TGA) (5% wt. loss)	330	330	°C	330	330	°C	IPC-TM-650.2.4.24.6
Pressure Cooker - 60 min then solder dip							IPC-TM-650.2.6.16
@288°C until failure (max 10 min.)	Pass	Pass		Pass	Pass		(modified)
T_{260}	16	16	minutes	16	16	minutes	IPC-TM-650.2.4.24.1
Chemical / Physical Properties							
Moisture Absorption0.07	0.07	wt. %	0.07	0.07	wt. %		IPC-TM-650.2.6.2.1
Methylene Chloride Resistance	0.31	0.31	% wt. chg.	0.31	0.31	% wt. chg.	IPC-TM-650.2.3.4.3
Density [50% resin content]	1.97	1.97	g / cm3	1.97	1.97	g / cm3	Internal Method

Park Electrochemical Corp. is a global advanced materials company which develops and manufactures high-technology digital and RF/microwave printed circuit materials and advanced composite materials, parts and assemblies. The company operates under the Nelco®, Nelcote® and Nova™ names.

All test data provided are typical values and not intended to be specification values. For review of critical specification tolerances, please contact a Nelco representative directly. Nelco reserves the right to change these typical values as a natural process of refining our testing equipment and techniques.

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*CAF resistance has been established to greater than 500 hours using a specific OEM coupon design and test procedure. For details on this or other CAF tests, please visit www.parkelectro.com.

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